# Assignment-1: Benchkit

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# 1. Intro

The benchmark chosen for porting to the updated Benchkit V2 campaign API is the `cuda_samples_matmul`. To start, context of the chosen benchmark will be discussed in Section 2. Modifications made will be discussed in Section 3, with use of a command wrapper in Section 4. Discussion of benchmark results will be done in Section 5.

# 2. Context

It is a matrix multiplication microbenchmark for testing the performance of performing matrix multiplication on a GPU's. It is tailored to test the performance of GPU's.

Specified in the benchmark variables are the dimensions of the matrix that are to multiplied with each other.

# 3. Bench

This section will briefly discuss the updated benchmark implementation;

## 3.1. Fetch

The fetch stage of the benchmark has been updated with cloning the completely example repository list in the original benchmarks documentation.

## 3.2. Build

The build stage is specially tailored to compiling and building the `matrixMul.cu` kernel at the moment. The `cmake` & `make` commands are sequentially executed for compiling said example.

## 3.3. Run

This stage executes a single program execution, existing bench code has been reused.

## 3.4. Collect

The existing collect stage has been copied and no modifications has been made to this stage code.

# 4. Command wrapper

The chosen wrapper for the updated benchmark is the `ncu2.py` wrapper, which is the cli version of NVIDIA Nsight Compute[1].

Using the GPU's hardware performance counters, more information from the execute kernel can be gained.

[1] https://developer.nvidia.com/nsight-compute

# 5. Analysis

This section will discuss analyzing the results of executing the updated benchmark for the Cuda matrix multiplication example.

## 5.1. Base

Included in the output of the base cuda kernel performance metrics are the following values:

- ma_width: Matrix A size.
- ma_height: Matrix A size.
- mb_width: Matrix B size.
- ma_width: Matrix A size.
- GFlops/s: Compute throughput per second.
- Time ms: Execution time.
- Ops: Total number of operations executed.
- Workgroup Size: Indicates the number of threads in a work group.

### 5.1.1. Graphs

## Bibliography